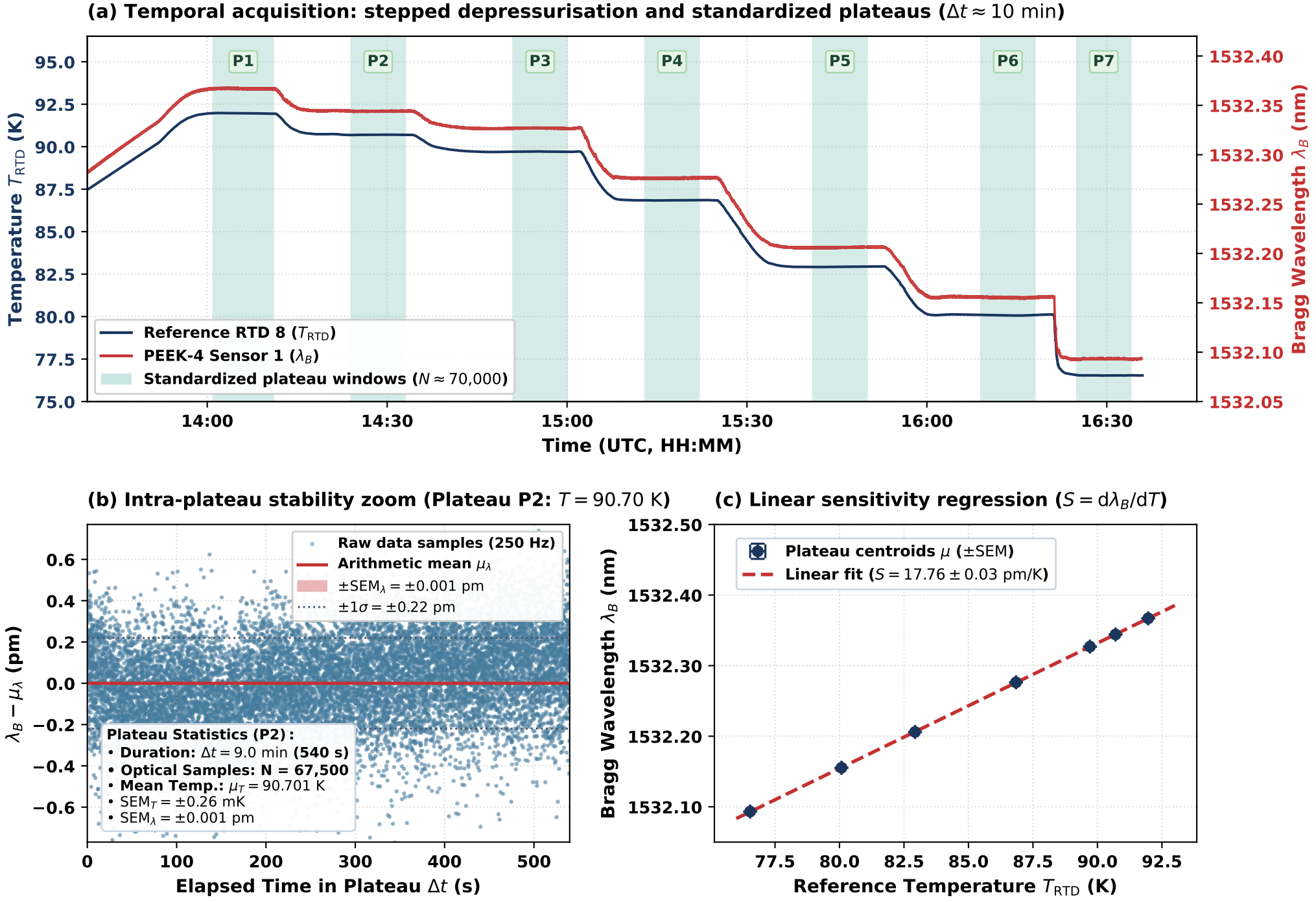

(a) Temporal acquisition: stepped depressurisation and standardized plateaus ($\Delta t \approx 10$ min)
P1
P2
P3
P4
P5
P6
P7
Temperature T_{RTD} (K)
Bragg Wavelength λ_B (nm)
Reference RTD 8 (T_{RTD})
PEEK-4 Sensor 1 (λ_B)
Standardized plateau windows ($N \approx 70{,}000$)
Time (UTC, HH:MM)
(b) Intra-plateau stability zoom (Plateau P2: $T = 90.70$ K)
Raw data samples (250 Hz)
Arithmetic mean μ_λ
$\pm SEM_\lambda = \pm 0.001$ pm
$\pm 1\sigma = \pm 0.22$ pm
Plateau Statistics (P2):
• Duration: $\Delta t = 9.0$ min (540 s)
• Optical Samples: N = 67,500
• Mean Temp.: $\mu_T = 90.701$ K
• $SEM_T = \pm 0.26$ mK
• $SEM_\lambda = \pm 0.001$ pm
$\lambda_B - \mu_\lambda$ (pm)
Elapsed Time in Plateau Δt (s)
(c) Linear sensitivity regression ($S = d\lambda_B/dT$)
Plateau centroids μ (±SEM)
Linear fit ($S = 17.76 \pm 0.03$ pm/K)
Bragg Wavelength λ_B (nm)
Reference Temperature T_{RTD} (K)